Titanic Data Workflow: Module Summary

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GitHub repository: https://github.com/iconrealestate77/ai-programming-foundations-project

1. Overview

This project is a reproducible data workflow in Python that loads, cleans, explores and visualizes the Titanic passenger dataset. The goal is to understand which passenger characteristics were associated with survival. The dataset is the seaborn-data copy of the Titanic data (891 passengers): https://github.com/mwaskom/seaborn-data/blob/master/titanic.csv. All work is in the notebook *data_workflow.ipynb*.

2. Dataset Description

The dataset records passengers of the Titanic, with one row per passenger. It has 891 rows and 15 columns before cleaning, and 13 columns after cleaning (the redundant *alive* column and the mostly missing *deck* column were dropped). The key variables are *survived* (0 or 1), *sex*, *pclass* (ticket class 1 to 3), *age* and *fare*. Before cleaning, *age* was missing for 177 passengers, *embarked* and *embark_town* for 2 each, and *deck* for 688 passengers. The data is a sample of the people aboard and not a complete passenger list.

3. Workflow Description

- **Ingestion:** *titanic.csv* is read with Pandas from the same folder as the notebook, and the first rows are displayed to confirm that it loaded.
- **Cleaning:** two documented functions, *standardize_columns* and *handle_missing_values*, standardize text labels, drop *alive* and *deck*, and fill missing values.
- **Exploratory analysis:** the function *survival_summary* reports passenger counts and survival rates by group, and *correlation_check* reports a correlation matrix for numeric columns.
- **Visualizations:** three Seaborn charts (Figures 1 to 3), each with a title, labeled axes and a written interpretation.
- **Summary:** a final markdown section describes findings, assumptions and limitations.

4. Key Decisions and Assumptions

Cleaning choices. Missing ages (about 20% of passengers) were filled with the median age of passengers who share the same sex and class, so that all 891 rows could be kept. The median was chosen because it is less affected by extreme values than the mean. The 2 missing port values were filled with the most common value, which is a very small assumption. The *deck* column was dropped because about 77% of it was missing, so any fill would be mostly invented. Text columns were lowercased and stripped of whitespace so that one category cannot be counted as two different labels.

Extreme fares were kept because they correspond to real first-class tickets. Wrapping each step in a function, instead of editing the data by hand, follows the advice to avoid manual data manipulation steps (Sandve et al., 2013).

EDA focus. I focused on survival by sex, class and age group, because these are the most natural passenger characteristics to compare, and I checked correlations to see whether fare and age add information beyond class.

What each plot shows. Figure 1 compares survival rates for women and men. Figure 2 adds passenger class, so that sex and class can be read together. Figure 3 shows the age distribution split by outcome, to see whether very young or very old passengers were treated differently.

Design decisions supported by sources. I kept the work in a single Jupyter notebook with clear section headings, small reusable functions with docstrings, and open-source Python tools, which matches the reproducible, transparent workflow promoted by the open learning resource of Danchev (2022). I also put the code under version control and recorded the package versions I used, which correspond to the practices of version controlling custom scripts and archiving the exact versions of external programs (Sandve et al., 2013).

5. Results and Interpretation

About 38% of the 891 passengers survived. Sex shows the largest difference: women survived at about 74% and men at about 19% (Figure 1). Survival also fell with class, from about 63% in first class to 47% in second class and 24% in third class. Figure 2 shows that class and sex work together: about 97% of first-class women survived, compared with about 14% of third-class men. The correlation table agrees with this: *survived* has a moderate negative correlation with *pclass* (-0.34), a weaker positive correlation with *fare* (0.26), and almost no linear correlation with *age* (-0.06). Even so, children aged 0 to 12 survived at about 58%, higher than any adult age group. In Figure 3 the youngest bars contain a larger share of survivors, while most passengers aged roughly 20 to 40 did not survive.

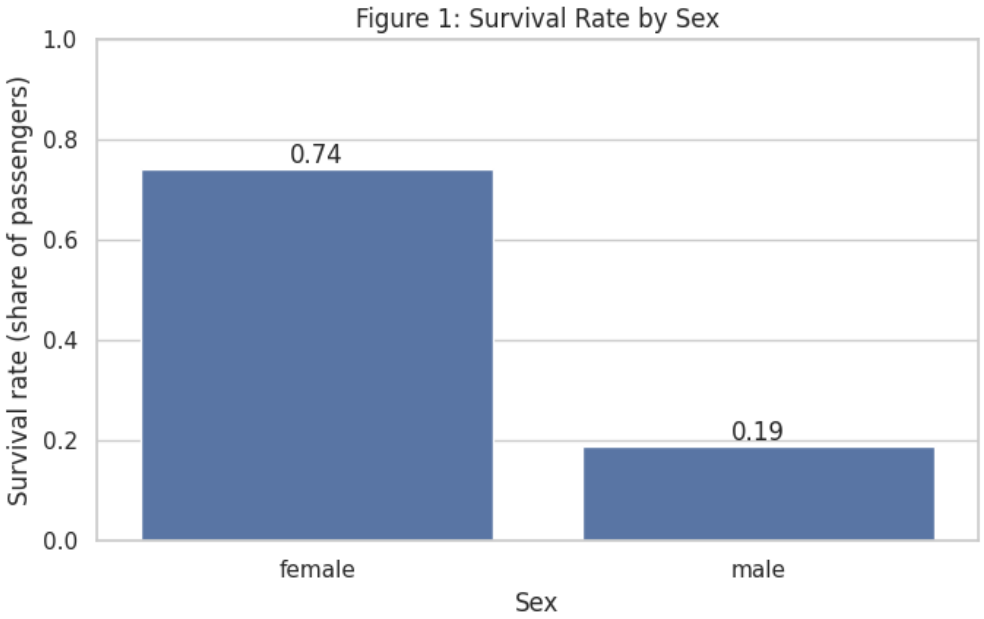


Figure 1. Survival rate by sex.

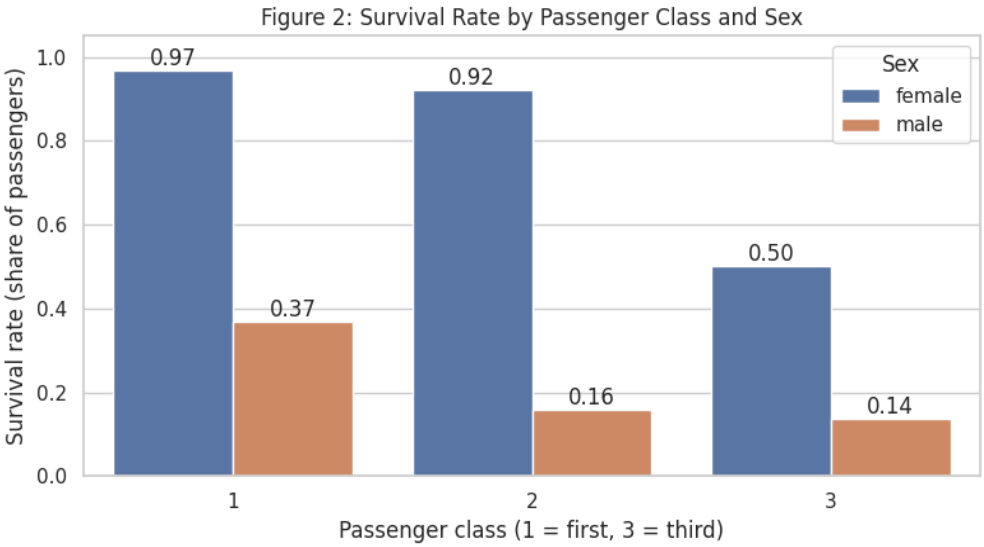


Figure 2. Survival rate by passenger class and sex.

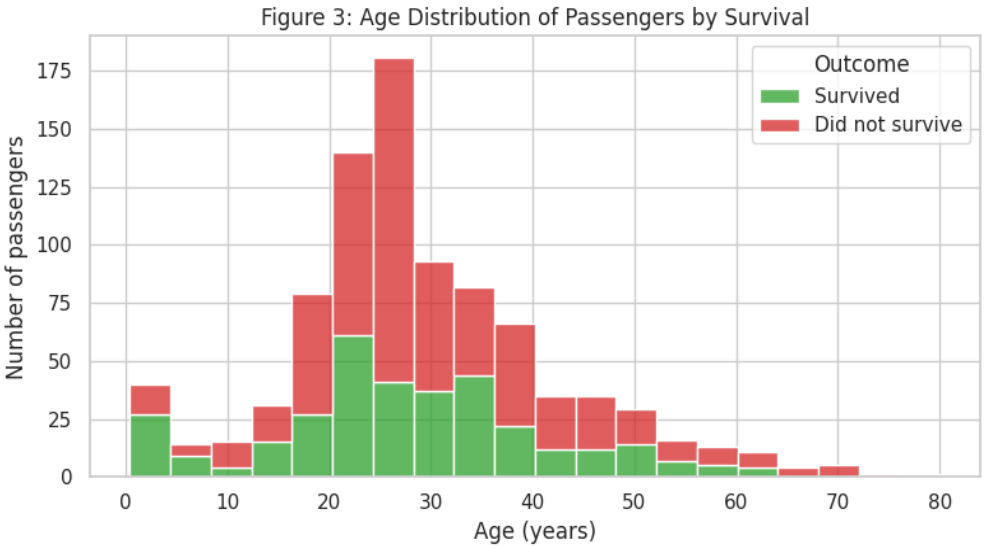


Figure 3. Age distribution of passengers by survival outcome.

These results are associations. The variables are related to one another (for example, first-class passengers paid higher fares), so the data cannot show how much each factor matters by itself or prove why the differences exist.

6. Responsible Practice (Bias and Data Quality)

The way missing values are handled can change the results. Ages were not missing evenly: about 28% of third-class passengers had no age, compared with about 14% in first class and 6% in second class. Deleting those rows would have under-represented third class, which had the lowest survival rate, and so would have pushed the survival estimates upward. Filling every missing age with one overall average would also be misleading, because it would give very different groups the same age. I used group medians instead, but this still shrinks the natural spread of ages and creates an artificial peak, which is visible in Figure 3. The dropped *deck* column also means nothing can be said about where passengers were on the ship. To reduce these risks I documented every cleaning step and its reason in the notebook, so a reader can judge it. Next I would compare the results with and without the filled ages and test a different imputation method to see whether the conclusions change.

7. Reproducibility

Someone else can rerun this work by cloning the repository, creating a virtual environment, running *pip install -r requirements.txt*, and opening *data_workflow.ipynb* with Kernel, Restart and Run All; *titanic.csv* must stay in the same folder as the notebook. The file *requirements.txt* was created with *pip freeze* from a dedicated virtual environment, so it lists the packages used in this project (for example, Python 3.8.2, pandas 2.0.3, NumPy 1.24.4, Matplotlib 3.7.5 and Seaborn 0.13.2). Work was committed step by step on a separate *dev* branch, with commit messages that name each task, so the progress of the project can be traced in the repository history. This follows the practice of recording how each result was produced and keeping scripts under version control (Sandve et al., 2013), and of building an open,

transparent workflow in Jupyter (Danchev, 2022).

8. References

Danchev, V. (2022). Reproducible Data Science with Python: An Open Learning Resource. *Journal of Open Source Education, 5*(56), 156. https://doi.org/10.21105/jose.00156

Sandve, G. K., Nekrutenko, A., Taylor, J., & Hovig, E. (2013). Ten simple rules for reproducible computational research. *PLoS Computational Biology, 9*(10), e1003285. https://doi.org/10.1371/journal.pcbi.1003285

Waskom, M. (n.d.). *seaborn-data: Titanic dataset* [Data set]. GitHub. https://github.com/mwaskom/seaborn-data/blob/master/titanic.csv